WhatsApp Chat Analyzer

Data Analysis and Visualization of WhatsApp Conversations

Team Members:

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- Himanshu
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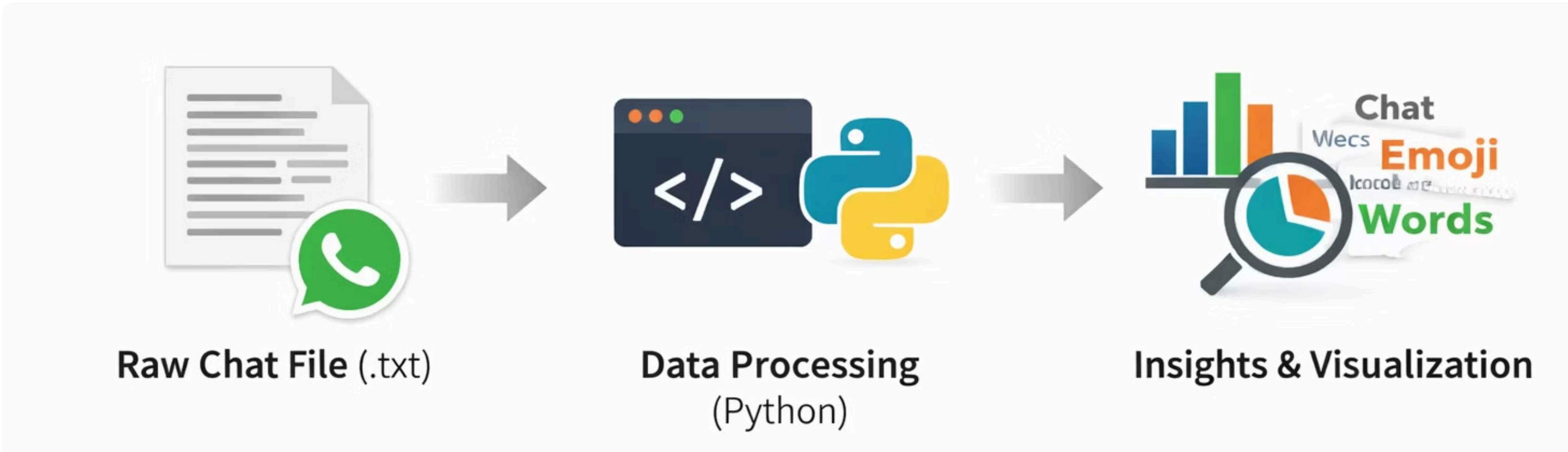
Problem Statement & Objectives



🔹 Section 1: Problem Statement

- WhatsApp generates large volumes of unstructured chat data.
- Manual analysis of conversations is difficult and time-consuming.
- No built-in advanced analytics in exported chat files.
- Need for extracting meaningful insights from textual conversations.

🔹 Section 2 : Project Goals

- Convert raw WhatsApp chat (.txt) into structured data.
- Perform statistical and textual analysis.
- Identify active users and conversation trends.
- Analyze most frequent words and emojis.
- Visualize insights using graphs and WordCloud.

Dataset & Technology Stack

Dataset Description

- Source: Exported WhatsApp chat (.txt file)
- Format: Semi-structured text
- Each message contains:
- Date
- Time
- Sender Name
- Message Content
- Includes media notifications and system messages

```
21/12/23, 16:10 - Akshat Gupta (Icfai): Call kar batata hu
21/12/23, 16:10 - Atul (Icfai): Continuous evaluation
21/12/23, 16:10 - Atul (Icfai): Me
```

Technology Stack

Python (Core Programming)

Pandas (Data Manipulation)

Plotly (Data Visualization)

Emoji Library (Emoji Extraction)

TextBlob (Sentiment Analysis)

WordCloud (Text Visualization)

Data Preprocessing

```
21/12/23, 14:11 - Indraneel (Icfai): Maths or computer ka bhi de diya
21/12/23, 14:28 - Akshat Gupta (Icfai): Batao bhai kitne kitnr aaye sabh 😋😋
21/12/23, 15:36 - +91 89389 00587: Koi link bhej do yr portal ki
21/12/23, 15:38 - Akshat Gupta (Icfai): https://pms.ibsindia.org/iudsz/
21/12/23, 15:53 - Bhaskar (Icfai): Mt dekh bhaiii
21/12/23, 15:53 - +91 89389 00587: Abe mil hi nahi Raha hai
21/12/23, 16:09 - Akshat Gupta (Icfai): Kya nahi mil raha tuje
21/12/23, 16:10 - +91 89389 00587: Result kaha hota hai
21/12/23, 16:10 - Akshat Gupta (Icfai): Call kar batata hu
21/12/23, 16:10 - Atul (Icfai): Continuous evaluation
21/12/23, 16:10 - Atul (Icfai): Me
21/12/23, 16:11 - Akshat Gupta (Icfai): Va bhai kya bat batao hae 🤣🤣🤣🤣🤣pehle wala step toh batata pehle academics par ja phir cont
nual evolved
21/12/23, 16:11 - Akshat Gupta (Icfai): This message was deleted
21/12/23, 16:11 - Akshat Gupta (Icfai): Evolution ***
21/12/23, 22:11 - +91 78589 11772: Kisko kisko back lga bhai
21/12/23, 22:11 - +91 78589 11772: ??
```

1

Extracting Date & Time

- Used Regular Expression to identify message pattern
- Regex pattern used:

```
r'\d{1,2}/\d{1,2}/\d{2,4},\s\d{1,2}:\d{2}\s-\s'
```

- Used `findall()` to extract date-time entries
- Used `split()` to separate messages

2

Cleaning Steps

- Removed `<Media omitted>`
- Removed system notifications
- Applied stopword filtering
- Standardized text format

3

Converting to Structured Format

- Created Pandas DataFrame
- Columns created:
- Date
- Time
- User
- Message

	date	user	message	year	month	day	hour	minute
0	2023-12-01 13:45:00	group_notification	Messages and calls are end-to-end encrypted. O...	2023	December	1	13	45
1	2023-09-01 21:54:00	group_notification	~ Shreedhar Sharma created group "B tech group...	2023	September	1	21	54
2	2023-12-01 13:45:00	group_notification	Akshat Gupta (Icfai) added you\n	2023	December	1	13	45
3	2023-12-02 22:25:00	Omkar Chaturbedi (Icfai)	<Media omitted>\n	2023	December	2	22	25
4	2023-12-02 22:25:00	Omkar Chaturbedi (Icfai)	<Media omitted>\n	2023	December	2	22	25
...	...	...	...	...	...	...	...	...
6230	2026-02-06 22:17:00	+91 96340 79709	Iske baki k kand secret hi rkhe h\n	2026	February	6	22	17
6231	2026-02-06 22:19:00	Kunal (Icfai)	epstein files ❌\nICFAI files ✅\n	2026	February	6	22	19
6232	2026-02-06 22:19:00	+91 96340 79709	😂\n	2026	February	6	22	19
6233	2026-02-06 22:21:00	Gauri Shankar (Icfai)	<Media omitted>\n	2026	February	6	22	21
6234	2026-02-06 23:48:00	+91 88648 73697	😂\n	2026	February	6	23	48

Basic Statistical Analysis

5K+

Total Messages

Aggregate message count across entire conversation history

85K+

Total Words

Cumulative word count excluding media and system messages

247

Media Messages

Images, videos, documents, and audio files shared

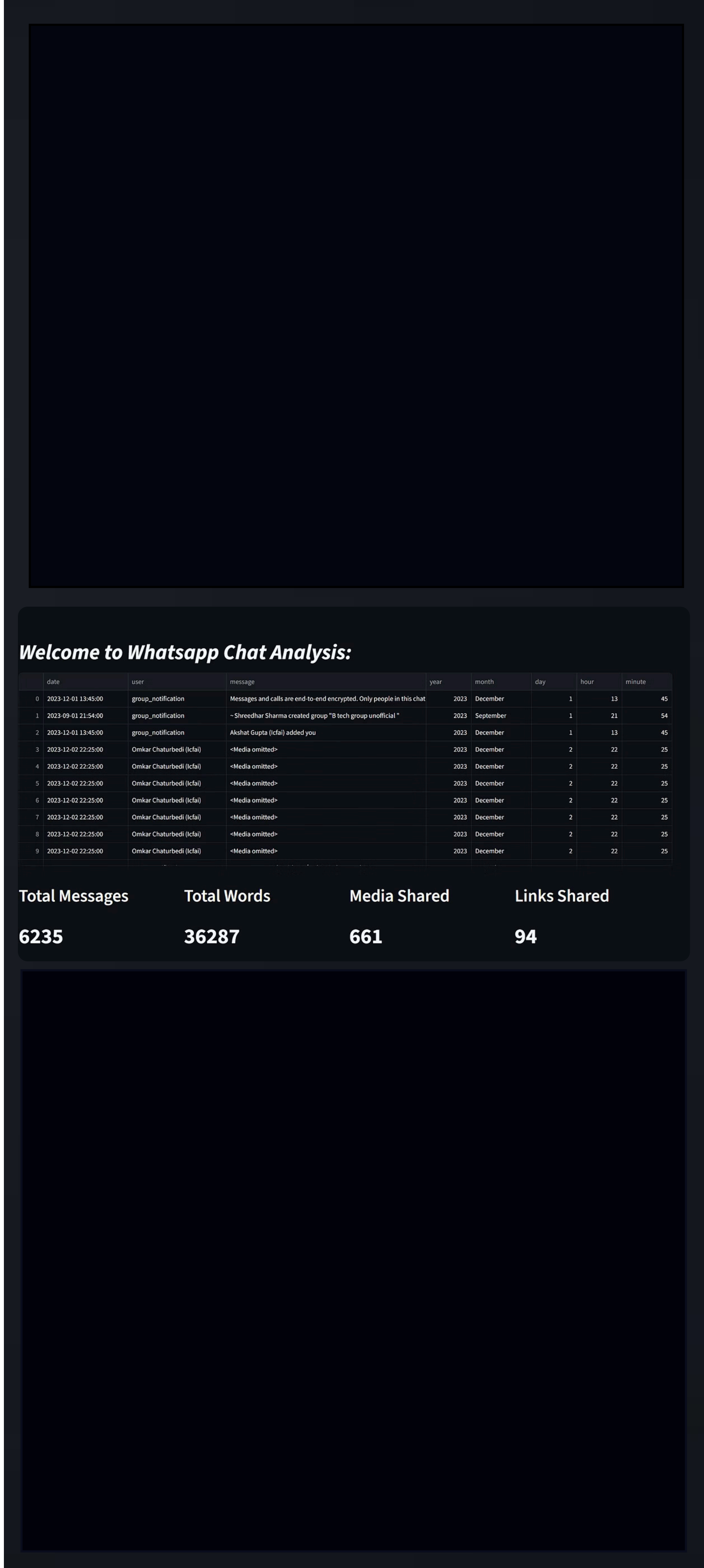
89

Links Shared

URLs and web links exchanged during conversations

📌 How It Was Calculated:

- `len(df)` → Total messages
- Word count → Split each message and count
- Media detection → Filter `<Media omitted>`
- Link detection → Using URLextract library



User Activity Analysis

1

📌 Objective

- Identify most active participants in the chat.
- Analyze message distribution among users.

2

📌 Method Used

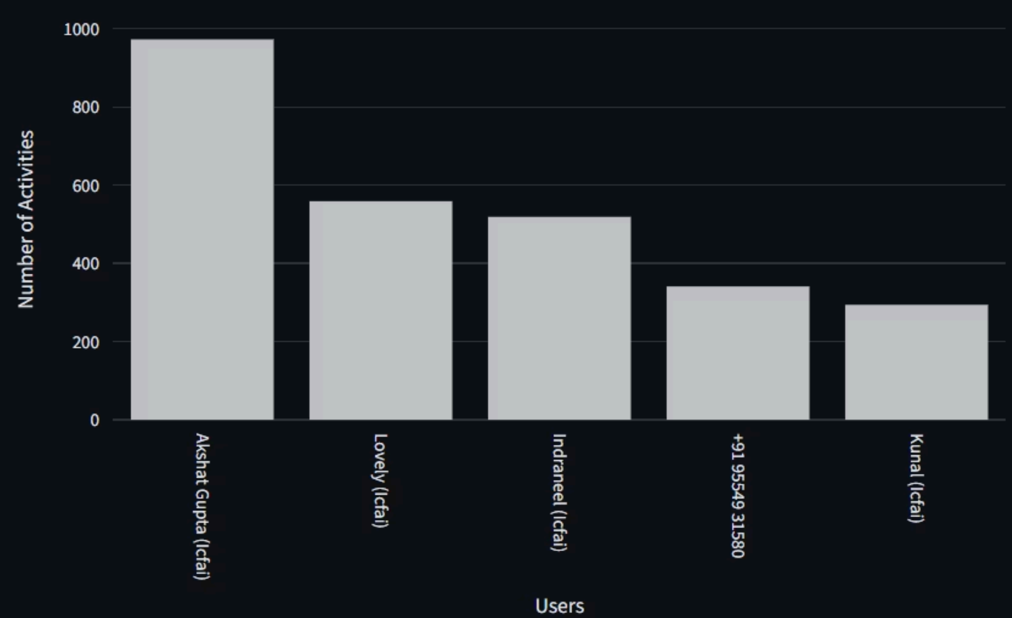
- Grouped DataFrame by `User`
- Counted number of messages per user
- Sorted in descending order
- Visualized using bar chart (Matplotlib)

3

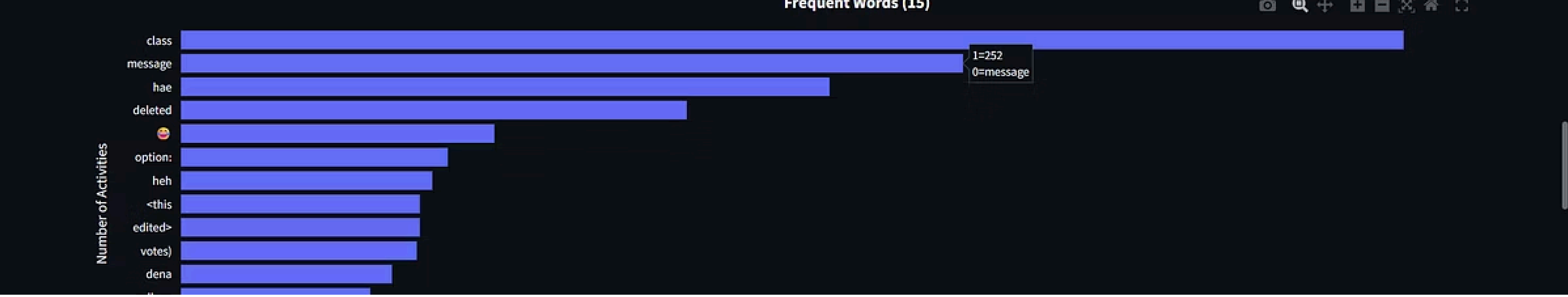
📌 Output

- Top active users
- Percentage contribution of each user

Most Busy Users



	Name	Percent
18	Anuj Mishra (Icfai)	1.69
19	Sahil	1.64
20	Aditya Singh (Icfai)	1.43
21	Lochan Naidu (Icfai)	1.13
22	+91 79837 31118	1.06
23	+91 82506 59851	1.05
24	Atul (Icfai)	1
25	Atul 2	0.78
26	Divyam(Icfai)	0.61
27	S A	0.51
28	Rohan	0.38



Most Common Words Analysis

1 Objective

- Identify most frequently used words.
- Understand dominant conversation patterns.

2 Processing Steps

- Converted text to lowercase.
- Tokenized messages using `.split()`.
- Removed stopwords.
- Counted frequency using `Counter`.
- Extracted top 15 words.

3 Output

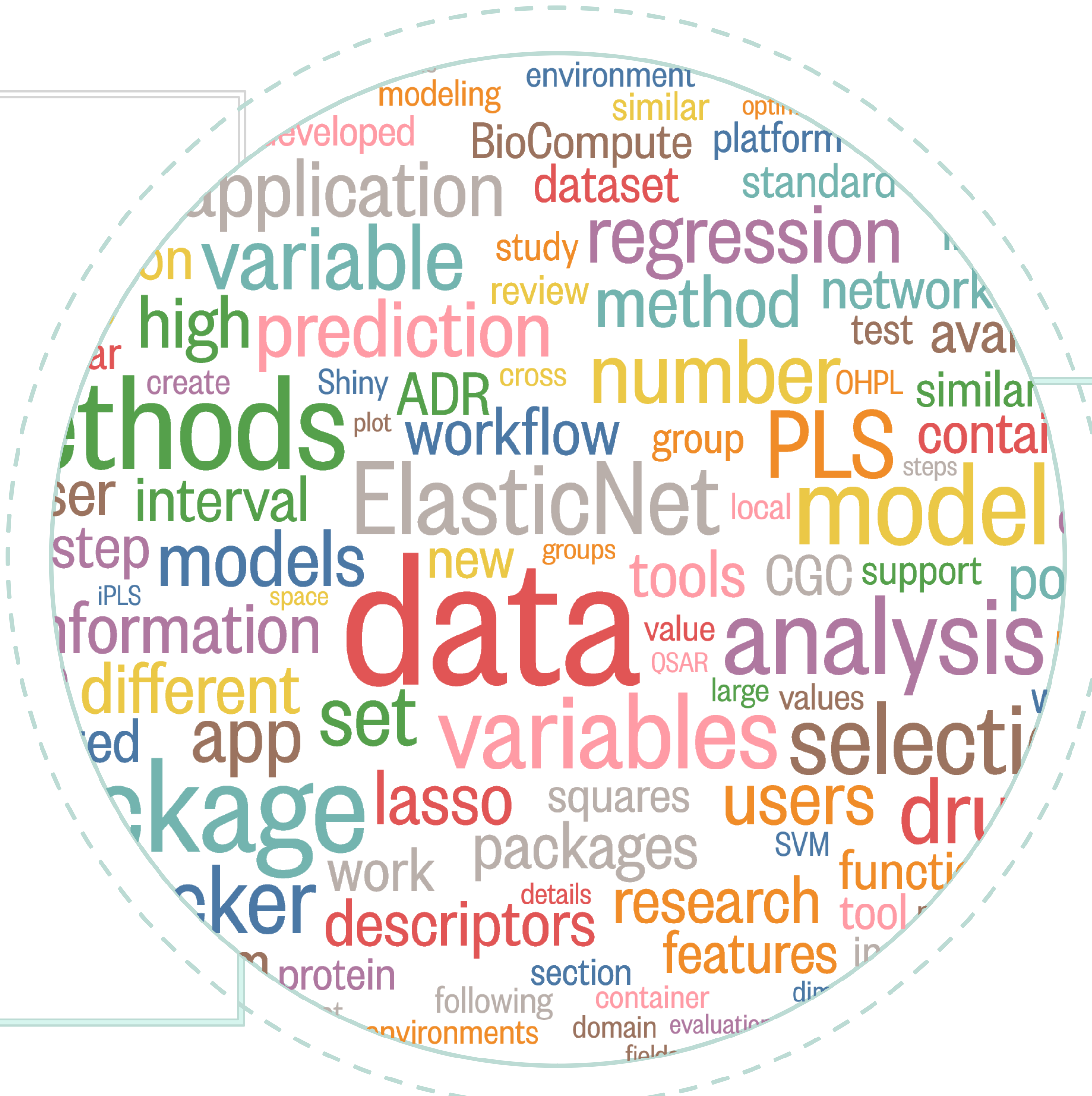
- Generated frequency table.
- Visualized using bar chart.

WordCloud Visualization

Objective

Provide visual representation of word frequency.

Identify dominant themes in conversation.



Methodology

Combined cleaned messages into single text corpus.

Removed stopwords and system messages.

Generated WordCloud using Python `WordCloud` library.

Rendered using Matplotlib.

Interpretation

Larger words → Higher frequency.

Smaller words → Lower frequency.

Helps quickly understand conversation focus.

Emoji Analysis

1	2	3

◆ Objective

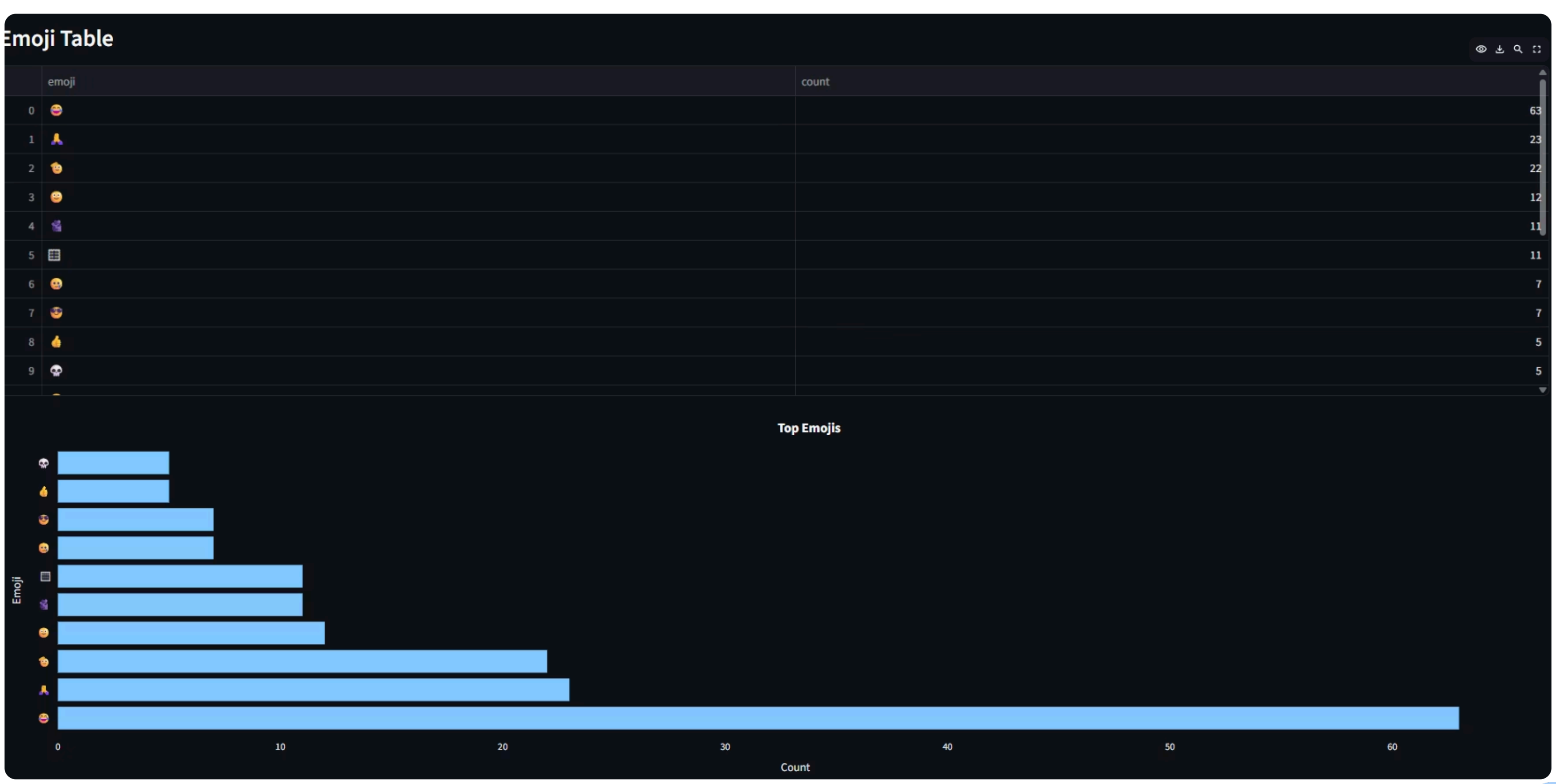
- Analyze emotional patterns in WhatsApp conversations.
- Identify most frequently used emojis.
- Compare emoji usage across users.

◆ Methodology

- Filtered data by selected user (or overall chat).
- Iterated through each message.
- Extracted emojis using `emoji.is_emoji()` (Unicode detection).
- Stored emojis in a list.
- Calculated frequency using `Counter`.
- Converted results into a Pandas DataFrame.
- Visualized top 10 emojis using a donut chart.

◆ Visualization

- Displayed emoji frequency table.
- Donut chart shows percentage contribution of top emojis.
- Inner radius improves readability and presentation clarity.



Activity Timeline Analysis

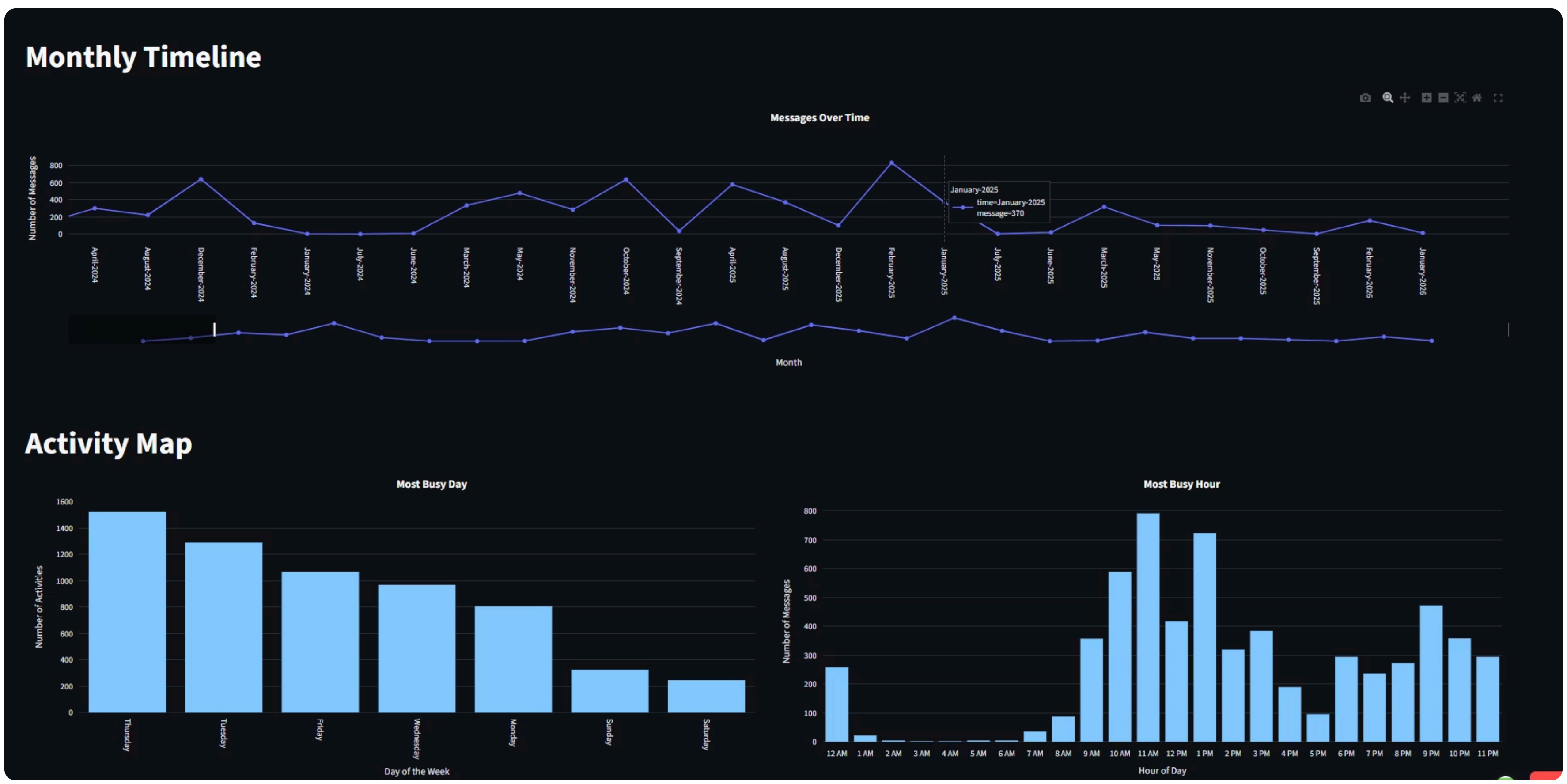
Monthly message trend visualization

Identifies most active day

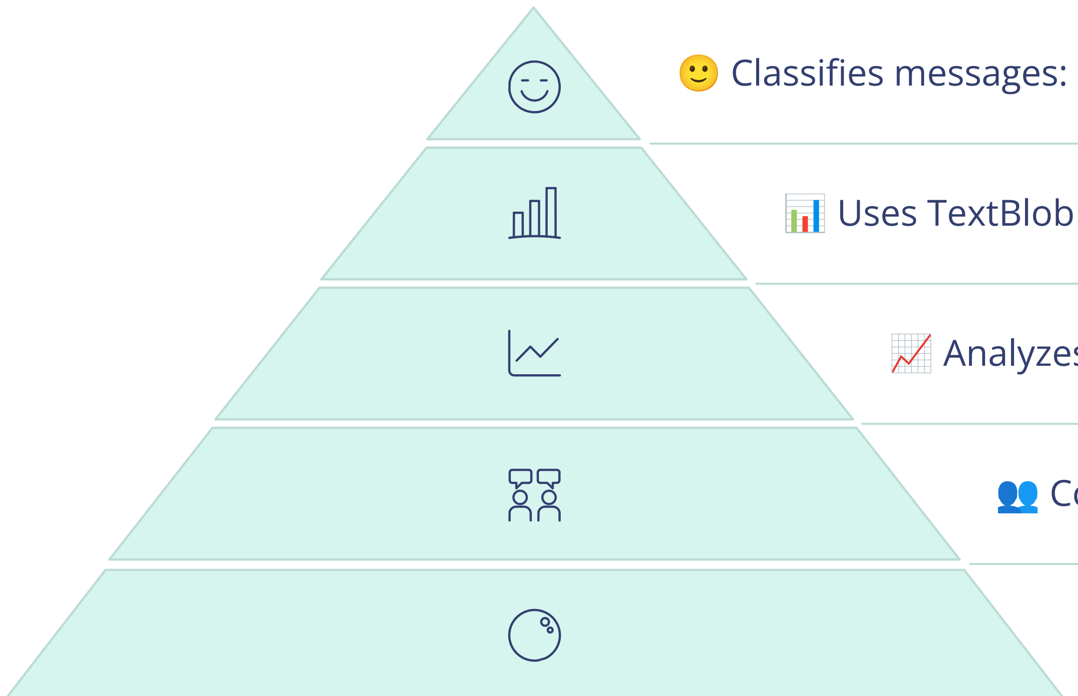
Detects peak chatting hours

Uses datetime-based grouping

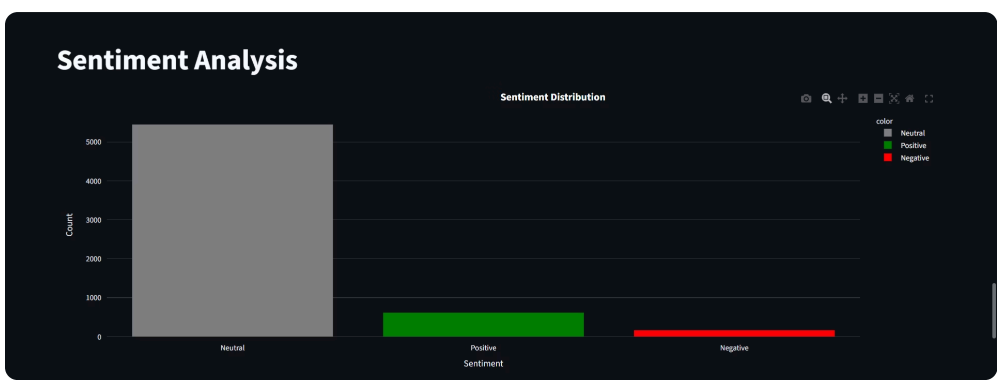
Reveals communication patterns

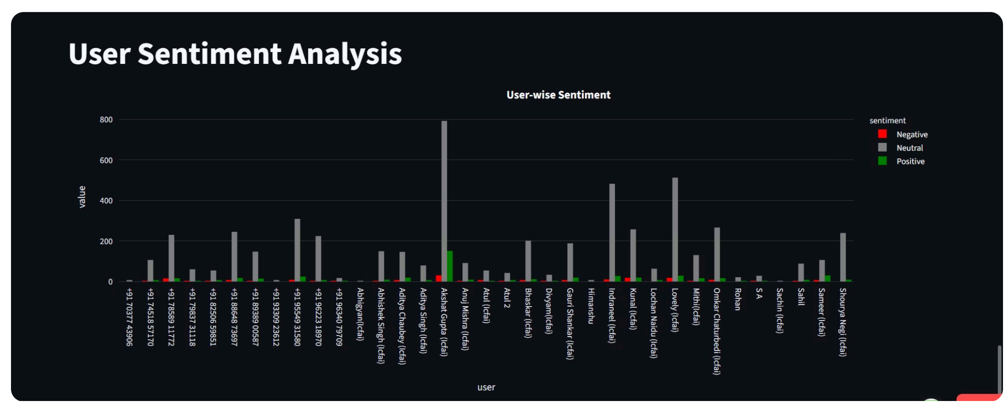


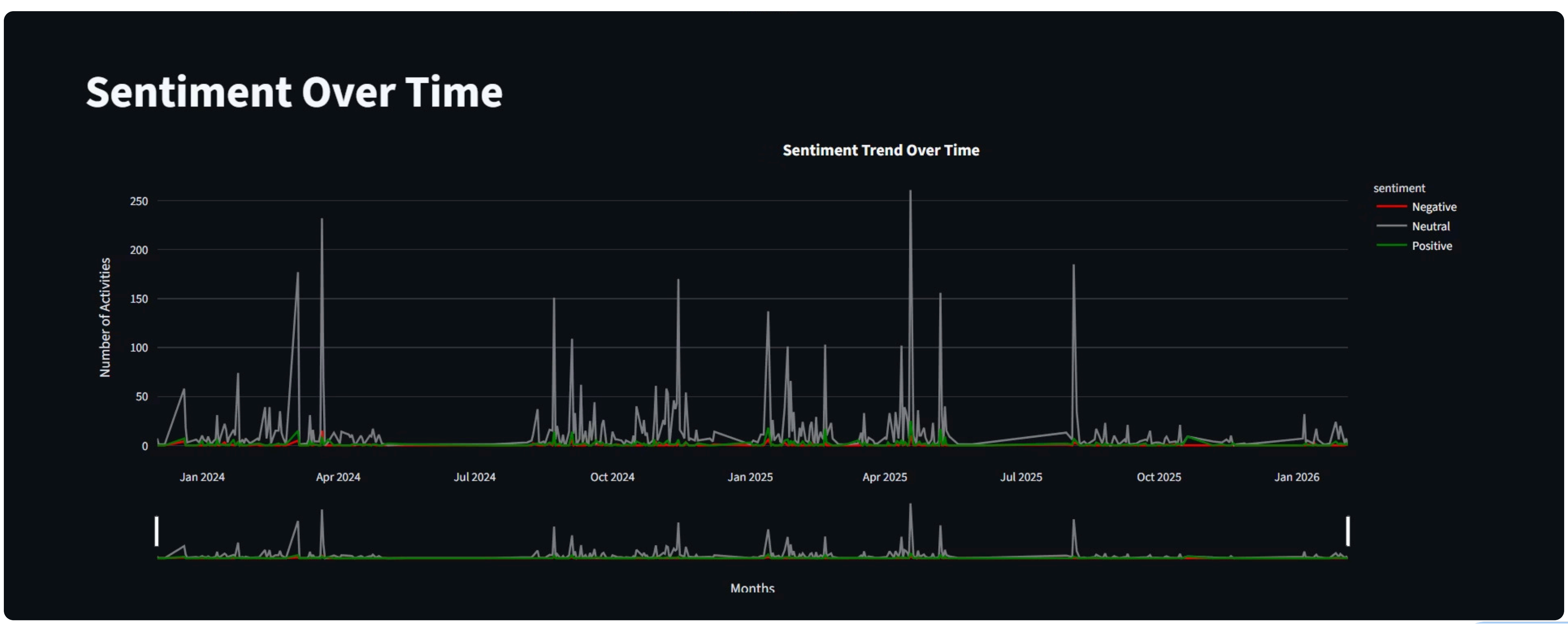
Sentiment Analysis



- 🙂 Classifies messages: positive, negative, neutral
- 📊 Uses TextBlob for sentiment scoring
- 📈 Analyzes sentiment distribution
- 👥 Compares sentiment across users
- 💭 Shows overall chat mood







user

Key Insights

Most Positive 😊	Most Negative 😡	Most Active 💬	Peak Day 📅
Anuj Mishra (Icfai)	S A	S A	Wednesday

Key Insights

Most active user identified

Most positive user detected

Peak chatting day identified

Most negative user detected

Conclusion

1	Handling multiple chat formats → Used flexible regex
2	Parsing multi-line messages → Merged lines correctly
3	Detecting emojis accurately → Used emoji library
4	Removing irrelevant words → Applied custom stopwords
5	Sentiment on Hinglish text → Used basic cleaning & thresholds

The WhatsApp Chat Analyzer successfully transforms unstructured chat data into meaningful insights. It helps in understanding user behavior, communication patterns, and overall sentiment of conversations. By combining data processing with visualization, the system provides an efficient and interactive way to analyze chat data, demonstrating practical application of data analytics techniques.

Thank You